

Enhancing the Programming Sequence for Undergraduate Computer Science Students: A Program for Improving Learning Outcomes

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Abstract— Arkansas Tech University (ATU) has over 650 students (2020-2021) from STEM (Science, Technology, Engineering, and Mathematics) majors taking programming sequence (PS) courses each semester. However, despite minor efforts to increase retention, the Engineering and Computing Sciences Department has had difficulty retaining students in the PS courses. The Programming Sequence Improvement Program (PSIP) has been implemented to address this issue. Its goal is to improve retention and student success in three PS courses: Foundations of Computer Programming I (COMS 2104), Foundations of Computer Programming II (COMS 2203), and Data Structures (COMS 2213). The PSIP aims to reduce the rate of D, F, Withdraw, and Incomplete (DFWI) grades by 10% annually from the current average baseline of 32%, 35%, and 27%, respectively (2015-2019). This program aims to revolutionize the Programming Sequence (PS) curriculum within the initial two years of study, fostering increased knowledge and improving overall degree completion rates. To achieve this goal, the PS Improvement Program (PSIP) employs four distinct approaches: establishing a dedicated tutoring lab for students enrolled in programming courses, integrating effective communication and collaboration tools like Discord and WebEx, conducting research on contemporary pedagogies for programming instruction, and aligning the curriculum with prominent platforms such as GitHub, Hacker Rank, and LeetCode. By implementing these strategies, the PSIP creates a transformative learning environment that enhances students' programming skills and facilitates their success in the program.

Keywords — stem, retention, computer programming, teaching pedagogies, curriculum.

I. INTRODUCTION

The Department of Engineering and Computing Sciences currently employs 11 full-time faculty members in three different major areas (Computer Sciences (CS), Information Technology (IT), and Cybersecurity (CSEC). The department has undertaken several ongoing projects to modernize its curriculum and faculty to ensure students enrolled in the program are exposed to the many facets of computer science, but more resources have hindered these efforts. Currently, at ATU, Programming sequence instruction COMS 2104 (Foundations of Computer Programming I), COMS 2203 (Foundations of Computer Programming II), and COMS 2213 (Data Structures) are available for the different majors within the computing and information science division. Improving these courses' sequential process, learning infrastructure, and retention significantly impact computer science degree and STEM completion rates.

The typical CS, CSEC, and IT students at ATU must master PS courses. Out of 661 students enrolled in different STEM majors in Fall 2021, CS students are the highest number (160) undertaking those courses. For many incoming first-generation college students, there is a cognitive conflict and gap that they experience. Many programming skills they learned in their K-12 training contradict the programming expected at the university level. For instance, the typical incoming first-year student who enrolls in a 2104 course is in the initial stages of the computational thinking process and has higher withdrawal rates are the target of this program. Still, later courses (2203 &

2213), which build on previous knowledge, had lower withdrawal rates in the 2020-2021 academic year. Also, on average, from 2015-2019, there was a slight increase in DFWI (D, F, Withdraw, and Incomplete) rates from COMS 2104 to COMS 2203; this is because of the complexity of the higher level of logical understanding. This program focuses on enhancing the PS curriculum.

II. LITERATURE REVIEW

Although there is a significant need for CS experts and many career opportunities, dropout rates are still high, and more is needed about what motivates people to finish their CS education and pursue a CS career. The U.S. has fallen from 1st to 16th in the world in the number of students completing college degrees [1]. According to studies on student attrition in computer science, most dropouts occur in the first two years of study [2]. According to the literature, 40% of students who enroll in CS degree programs eventually drop out without receiving a degree [3]. Student retention is influenced favorably by a degree's elevated level of utility. Additionally, the usefulness of a degree is positively impacted by a supportive atmosphere and cognitive improvements but negatively affected by non-cognitive gains [4]. Many STEM courses have been taught using a traditional, lecture-based approach, particularly in extensive introductory courses [5]. Compared to lecture-based methods, incorporating active-learning techniques into STEM classes has improved student learning and performance, boosted knowledge retention, and closed achievement disparities between diverse student populations [6]. In addition, computer instructors should incorporate creative, learner-centered teaching methods into their instructional process as they increase student engagement. Using innovative pedagogy enhanced undergraduate students' academic achievement and retention in computer programming [7].

III. METHODOLOGY

A. OBJECTIVE

Our objective for Programming Sequence Improvement Program (PSIP) was to increase the first-year and second-year student retention in Programming Sequence courses COMS 2104, COMS 2203, and COMS 2213 by decreasing DFWI ((D, F, Withdraw, and Incomplete grades) rates 10% each year, from the current average baseline of 32%, 35% and 27% respectively (2015-2019).

B. ACTION PLANS

The high-level goal of this PSIP was to change the face of the Programming Sequence (PS) program within the first two years of studies by enhancing programming techniques for students. The program's interventions were expected to significantly impact student achievement and retention in several STEM disciplines at ATU, resulting in increased degree completion rates. Each objective lends guidance to the overall successful

outcome of the proposed program. For example, each goal was intended to promote activities to students taking part in Programming Sequence courses with the specific work of preparing students in their STEM field of study by leveraging programming concepts and Data structures.

Action Plan 1: Creating comprehensive programming tutoring laboratories connected to the computer science (CS) curriculum within the Division of Computer and Information Science (CIS) to support dynamic, collaborative learning and project-based coding activities.

- a. Semester 1 (Spring 2022): Equipment acquisition and start deployment and utilization by at least 10~25 using them daily.
- b. Semester 2 (Fall 2022): Fully operational lab utilized by all ATU students enrolled in PS courses.
- c. Semester 3 (Spring 2023): Retention/completion in COMS 2104, COMS 2203, and COMS 2213 increased by 10%.
- d. Semester 4 (Fall 2023): Retention/completion in COMS 2104, COMS 2203, and COMS 2213 increased by 10% from semesters 1 & 2.

Action Plan 2: Developing a Social Media PS reaction system (SMPSRM): To measure students' understanding of programming/Data structure topics using social analytics tools in the classroom.

- a. Semester 2 (Fall 2022): 20~25 participants enrolled in COMS 2104, COMS 2203, and COMS 2213.
- b. Semester 3 (Spring 2023): 10% increase in participation from semester 2 in social media sites.
- c. Semester 4 (Fall 2023): 10% increase in participation from semester 3 in social media sites.

Action Plan 3: Programming Sequence Curriculum Alignment

- a. Identify curriculum content for which an increased emphasis on critical thinking and analytical skills is appropriate within several sections in the PS.
- b. Examine the creation of team-taught PS courses within CIS faculty members using lab technologies.
- c. Increase dialogue among faculty within the departments to increase awareness of relevance, usage, benefits of the changes within other disciplines, and the value of these approaches to STEM students' career paths.

Action Plan 4: Programming Sequence Curriculum Alignment Faculty Development in New Programming Sequence Pedagogies.

- a. Develop a training program for CIS faculty in PS pedagogies models for the PS curriculum.
- b. Develop a departmental wiki training series for faculty to train them to leverage digital tools such as Leet Code & Hacker Rank.
- c. Encourage faculty to participate in national conferences to receive training in the new curriculum efforts and student learning approaches.

IV. RESULTS

In 2019 – 2021 the DFWI rate of students taking programming courses for the first time was over 30%, around the same rate over the last six years. Based on the pandemic-related uncertainties, when many of us had to handle the overall student enrollment and retention, we excluded the 2019-2020 and 2020 – 2021 data points. Also, we believe that teachers and administrations were more lenient and forgiving during the pandemic years, and those years' data do not represent the regular DFWI rates for those courses. Hence, we decided to take the average DFWI rates of 2015- 2019, which are 32%,35%, and 27% in PS courses COMS 2104, COMS 2203, and COMS 2213, respectively.

Figures 1, 2, and 3 List first and second-year students' DFWI (D, F, Withdraw and Incomplete rates of COMS 2104, COMS 2203, and COMS 2213.

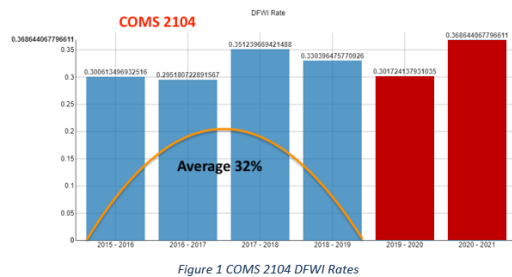


Figure 1 COMS 2104 DFWI Rates

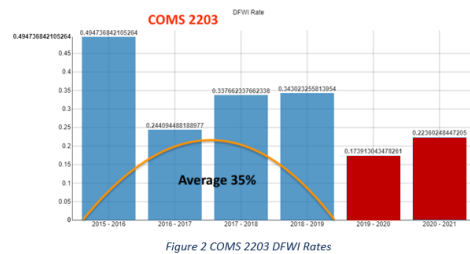


Figure 2 COMS 2203 DFWI Rates

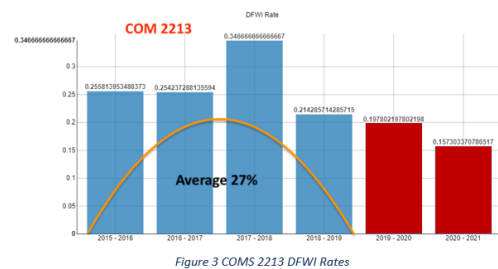


Figure 3 COMS 2213 DFWI Rates

A. EVALUATION PLAN

For the evaluation plan, we developed assessment tools to evaluate new educational strategies implemented within the community.

These are:

- Develop assessment tools for action plans 1, 2, and 3.

- Assess the impact of curriculum changes using tools developed in (a).
- Assess the ease of curriculum adoption using tools developed in (a).
- Administer a modified Motivated Strategies for Learning (MSL) questionnaire to measure the nature of students' motivations and usage of learning strategies related to the project [8].
- Administer classroom community scale questionnaire to students to measure their sense of community within their learning environments.
- Submit findings and data to conferences and journals that indicate project assessment results.

The program evaluation was done after each semester in four phases. We completed our activities for each step and evaluated the program's success. Since the program's duration is two years, we had three semesters to see the effects of the proposed activities. In addition, we are evaluating three classes (COMS 2104, COMS 2203, and COMS 2213) with the evaluation mentioned above plan.

B. STUDENT ENGAGEMENT SURVEY RESULT

From 2015-2019 average DFWI rate for those PS courses was 32%, 35%, and 27%, and our target was to reduce the 10% DFWI rate by Fall 2022. We have taken many initiatives to accomplish our goal: we have created a tutoring lab for students and implemented communication and collaboration tools like Discord and Webex. We have aligned our curriculum and familiarized the students with platforms like GitHub and Hacker Rank. Developing training programs for faculties also encourage faculties to participate in national conferences to receive training in the new curriculum. The DFWI rates for COMS 2104 (Programming I), COMS 2203 (Programming II), and COMS 2213 (Data Structures) have been successfully reduced from 32% to 23%, 35% to 15%, and 27% to 0% for the fall 2022 semester.

Figures 4, 5, and 6 list changes in DFWI (D, F, Withdraw, and Incomplete) rates of COMS 2104, COMS 2203, and COMS 2213 through Spring 2022 and Fall 2022.

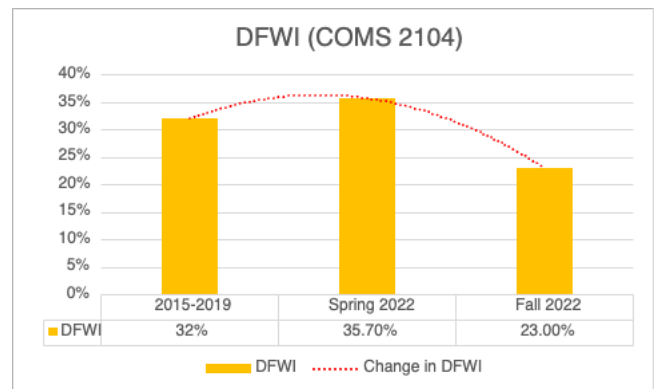


Figure 4 Change in DFWI COMS 2104

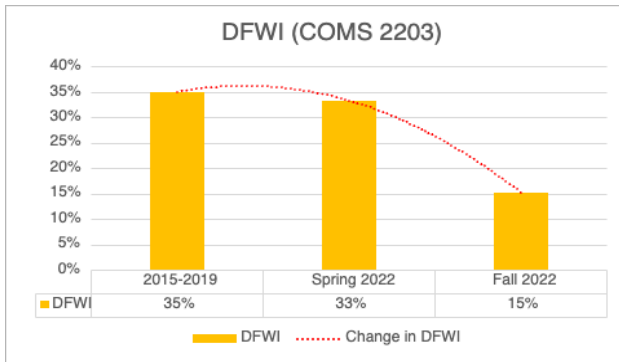


Figure 5 Change in DFWI COMS 2203

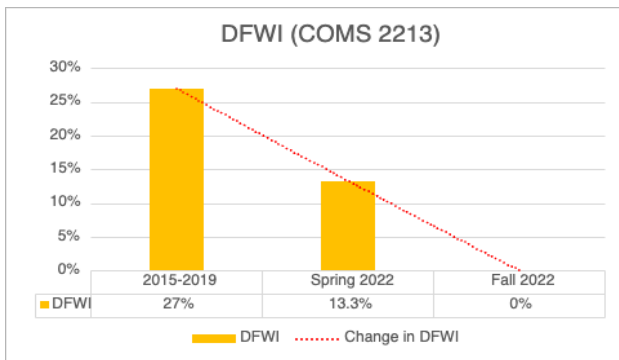


Figure 6 Change in DFWI COMS 2213

The student survey results show how they use platforms like GitHub, Discord, Leet Code, and Hacker Rank, which we have introduced for those PS courses, including our tutoring laboratories. Our tutoring labs offer in-person, Discord, and WebEx assistance to the students. Participants in the poll include 40 students for Programming I, 22 for Programming II, and 18 for Data Structures.

Figures 7, 8, and 9 list COMS 2104, COMS 2203, and COMS 2213 student survey results for Fall 2022.

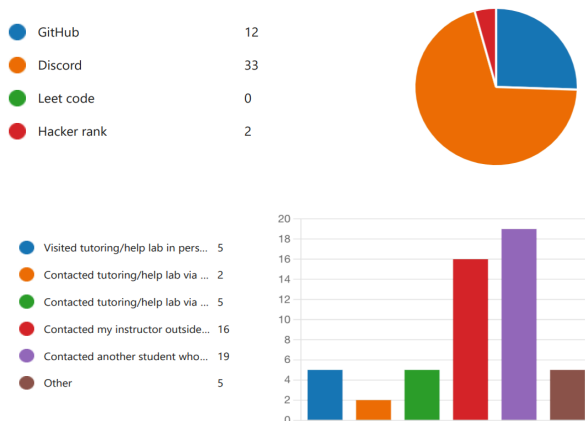


Figure 7 Students Survey Results in COMS 2104

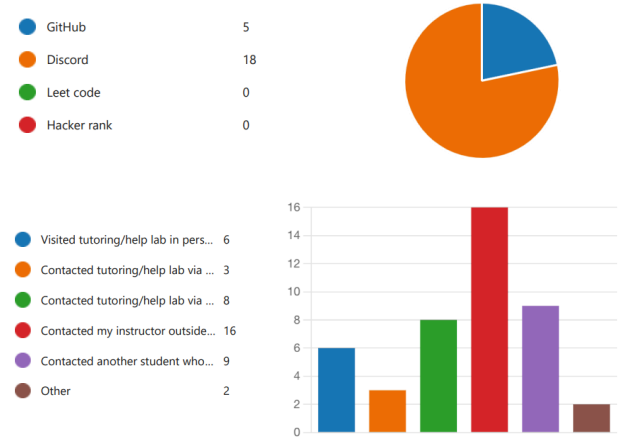


Figure 8 Students Survey Results in COMS 2203

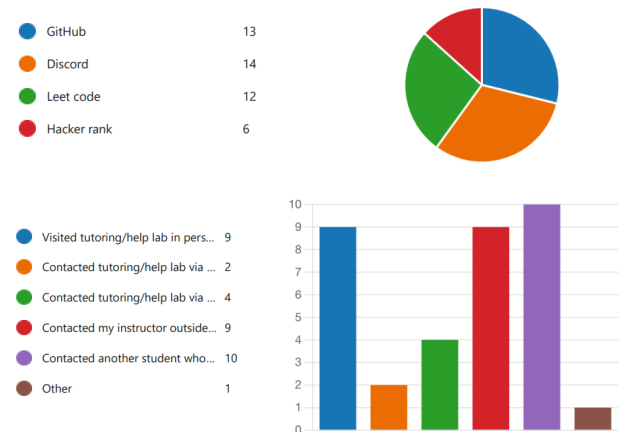


Figure 9 Students Survey Results in COMS 2213

V. DISCUSSION

This initiative is solely student-centric. The department currently manages all PS courses, which serve many STEM students. The courses combine the foundational concepts of programming using structured programming concepts of C++ as an implementation tool. By studying previous data, we realized that three different courses of programming sequence require special attention. As mentioned before, many students were affected positively by this program. First, with the curriculum alignment implemented in these courses, the students are getting similar standards regardless of the section or the faculty teaching that section. The student feedback on the preset assignments is a unique way of assessing their problems in these courses. The labs are also helping us identify the problem areas in these courses. The student workers who have already passed these courses have been able to help. We improved these three programming sequence courses by ensuring all the sections were coordinated and carefully

monitored. Our tutoring/help lab helps students both in-person and online. They receive guidance from the teacher outside the classroom and other students' assistance. That eventually helped us lower the DFWI rates in several STEM disciplines.

VI. CONCLUSION

In this study, we have implemented a comprehensive program to enhance the completion rates of the Programming Sequence (PS) courses for undergraduate computer science students. Our primary objective was to provide students with various assistance throughout these courses. By Fall 2022, our program aimed to reduce the DFWI rate by 10%. COMS 2104, COMS 2203, and COMS 2213 exhibited DFWI rates of 23%, 15%, and 0% during that semester. By successfully implementing multiple strategies, we could decrease the DFWI rates effectively. While the complete elimination of the DFWI rate remains challenging, our ongoing goal is to reduce the average below 10% further.

ACKNOWLEDGMENT

Arkansas Tech Innovation Grant,
Department of Engineering and Computing Sciences,
Arkansas Tech University, Russellville, Arkansas

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